ZP-M Illustrated Companion

THE ZERO PARADOX
ZP-M: Kleene–Ordinal Bridge

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This companion explains the ideas in plain language. It is not the formal document — every claim here restates a result already proved in ZP-M: Kleene-Ordinal Bridge. Consult that document for the authoritative mathematics.

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1. What This Layer Does

ZP-M is a consolidation layer. Its job is to connect two results that were proved separately in ZP-K and ZP-L, through maps between their types rather than an identity across them.

ZP-K names a code beside the initial state c_0 (⊥ of MachinePhase) as a KleeneStructure requirement (botCode_is_quine), a periodicity condition that constant codes also meet. Reading c_0 as the Kleene quine, a program that is its own program, splits in two. It is not an equation, since a code and a machine state are different types, so that half is retired; that ⊥ of MachinePhase runs is the occurrence commitment, and if it runs, nothing outside it runs it (DA-1, Path 1). ZP-L proved that nothing fires below the ordinal ε_0: the tower ω, ω^ω, ω^ω^ω, … approaches ε_0 from below, and any monotone map that sends tower stages to c_0 also sends every ordinal below ε_0 to c_0 (snap_threshold_is_epsilon_zero). Firing at ε_0 itself is not forced (the map that is c_0 everywhere meets both conditions); it is the hypothesis $h\varepsilon_0$, that the map sends ε_0 to c_1 (ZP-M §II), and given it, ε_0 is the least firing point. ZP-L also showed that the tower images in $\mathbb{Z}_2$ converge to 0.

ZP-M builds the bridge connecting these: a formal map snapEmbed that sends c_1 (the snap state) to 0 in $\mathbb{Z}_2$, and proves that all three objects — the ordinal ε_0, the snap state c_1, and the 2-adic zero — are formally co-witnessed in a single theorem (zpm_triangle). It also closes a gap in ZP-L's minimality theorem by showing that its free hypothesis follows from simpler conditions.

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2. The Type Bridge (snapEmbed)

The two layers — ZP-E's machine state space $\{c_0, c_1\}$ and ZP-B's 2-adic integers $\mathbb{Z}_2$ — are mathematically unrelated types. One is a two-element set; the other is an infinite algebraic structure. snapEmbed is the map that connects them structurally.

snapEmbed
c_0 (pre-snap) $\mapsto 1 \in \mathbb{Z}_2$ (a 2-adic unit, valuation 0)
c_1 (snap state) $\mapsto 0 \in \mathbb{Z}_2$ (the 2-adic zero, infinite valuation)

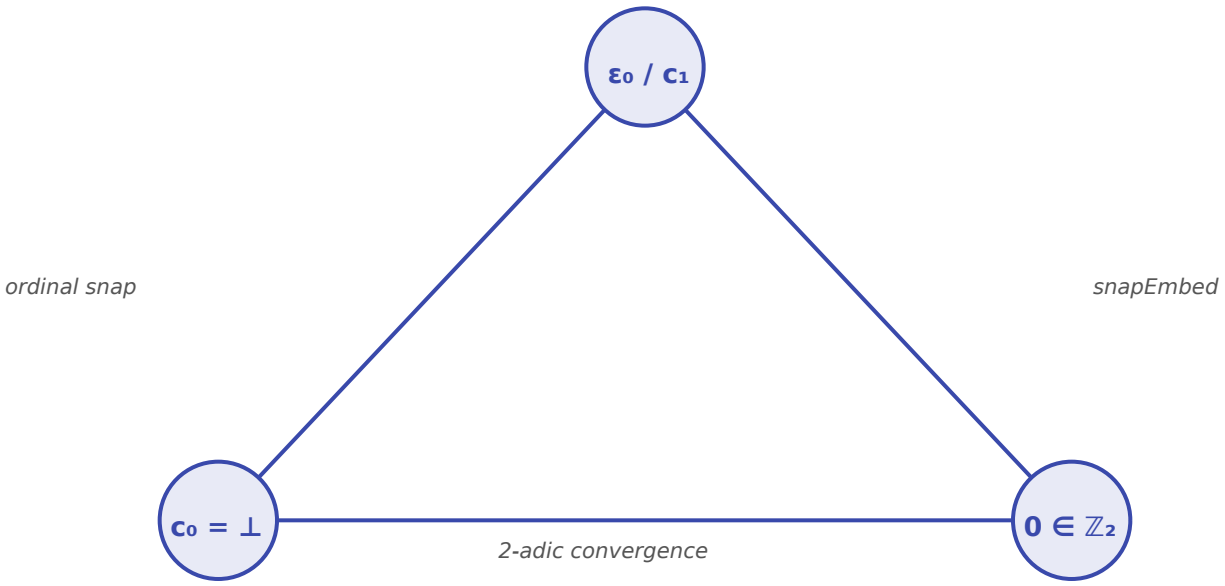
The map is injective (c_0 and c_1 land in distinct places) and preserves the algebraic structure: joining c_1 with anything gives c_1, and multiplying 0 by anything gives 0. In the die (ZeroParadox/Die.lean), with join as the step on the machine states and multiplication as the step on $\mathbb{Z}_2$, each is the only point that steps to itself and to nothing else, and snapEmbed is a map between those two motions (snapEmbedHom, ZeroParadox/Ordinal/Incompleteness.lean §I-b). Along it the role carries over at c_1, because 0 has only itself as a successor (snapEmbed_selfOnlyAt_c1), though the map's back clause fails at c_0, so the general transport does not apply. That relates the two occupants by a map; it does not make them one object. Under a self-application map supplied by AbstractSelfApp, the occupant on the machine states is c_0, by that class's fields: which point fills the role depends on the step.

The key geometric fact: 0 in $\mathbb{Z}_2$ is divisible by every power of 2. That is what "infinite 2-adic valuation" means, and under doubling 0 is the only point that steps to itself and to nothing else (z2_double_selfOnlyAt_iff). The snap step itself does not carry over to doubling: $c_0 \to c_1$ goes to $1 \to 0$, and 0 is not $2 \cdot 1$ (snapEmbed_doubling_noStepAt).

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3. The Triangle

Three objects that appear in three different layers of the framework are co-witnessed in a single Lean theorem: zpm_triangle.



The Kleene–Ordinal–2-adic triangle: three edges, one formal proof.

The left-side edge is the ordinal snap: below ε_0, the tower stages map to c_0; at ε_0, the canonical map flips to c_1. The right-side edge is the type bridge: snapEmbed sends c_1 to 0. The bottom edge is the 2-adic convergence proved in ZP-L: the tower images in $\mathbb{Z}_2$ converge to 0 as the ordinal stages approach ε_0.

All three edges are theorems, and zpm_triangle assembles them into a single formal statement. The ordinal, machine-state and 2-adic threads appear in the same proof; it has no computational conjunct, and the

Kleene reading of c_0 is not part of it.

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4. Closing a Gap in ZP-L

ZP-L's central minimality theorem (snap_exactly_at_epsilon_zero) carried a free hypothesis called hfp. It asserted that for any map from ordinals to machine phases, every fixed point of ω-exponentiation must be sent to c_1. This was not proved from the ordinal structure — it was an extra assumption.

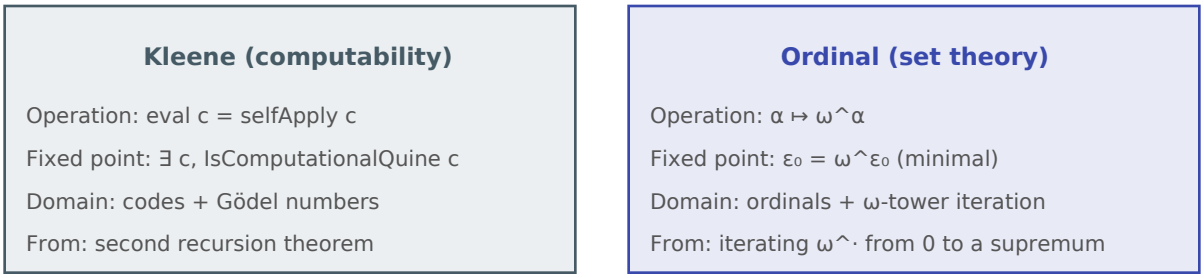
ZP-M §II closes this gap. The argument is short: if a map is monotone and sends ε_0 to c_1, then for any other fixed point $\alpha \geq \varepsilon_0$, monotonicity forces the map to agree with c_1 at α (because c_1 absorbs every join). The minimality theorem now holds under weaker hypotheses: monotonicity, every tower stage sent to c_0, and "snap at ε_0" together imply the full result (snap_unconditional), without hfp as a separate commitment.

> hfp is not an additional assumption — it is a consequence of monotonicity plus the alignment condition $\varphi(\varepsilon_0) = c_1$ (hfp_from_epsilon_zero). The gap in ZP-L is closed.

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5. Two Fixed Points, Two Domains

ZP-M closes by putting two fixed points side by side: the one Kleene's second recursion theorem gives (computability) and ε_0 (ordinal theory). Reading them as one argument running in two mathematical worlds is the framework's interpretation; the theorem proves less, as below.



Two fixed points in two domains, by two constructions, with no shared machinery.

In computability theory, a diagonal argument over a program's own Gödel number (its address in the code space) gives a code fixed up to evaluation. In ordinal theory, iterating ω-exponentiation from 0 and taking the supremum gives ε_0, the first ordinal the tower cannot exceed: the fixed-point lemma for normal functions, not a diagonal argument. The mechanisms are entirely different.

The theorem both_fixed_points_exist records the ordinal fixed point and, on the computability side, a code periodic with period its own Gödel number (constant codes also meet that condition, so it does not by itself pin self-reference), and the proofs live in the same Lean context. It is a conjunction of two existence proofs over different types, a Code and an Ordinal, so it does not show they are one pattern; no equation between the two is even well-formed.

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6. Remark R-M.1: Where the Frame Ends

ZP-M sets two of the three threads in the DA-1 argument beside the ordinal result. Path 1 (AFA structural) and Path 3 (incompressibility, whose Lean counterpart is a Kleene field) had their Lean counterparts carried together in ZP-K as a conjunction (da1_paths_unified), and Path 3's counterpart is not a witness of execution; that they are one structural fact, and that they share a frame with the ordinal result, is the framework's reading.

Path 2 (informational) remains outside this frame. It has two charts. Depth chart, ZP-C L-INF: surprisal grows without bound with depth in the ball hierarchy, each value finite (l_inf). Point chart: at 0 of $\mathbb{Q}_2$, the one point in every ball of the chain centred at 0, the 2-adic additive valuation is $\top$ (padic_addVal_bot), the point to which ZP-C has c_0, $\bot$ of the machine-state type, correspond. (As of 2026-11-01 no single declared map from the machine-state type into the 2-adic integers or numbers sends c_0 to 0: a search of the corpus's Lean files for both type names finds snapEmbed and two negative examples. A composite of two maps the corpus has does send c_0 to 0 of $\mathbb{Q}_2$: an order isomorphism onto a two-point type, read there as order shape only, then poleToSphere; that composite is not declared. snapEmbed, above, is a different map, sending c_0 to 1, and the two are not composed.) This is structurally analogous to the Kleene fixed point, but the analogy has not been made formal.

The reason, visible from ZP-M, is a framework separation: L-INF is a measure-theoretic result (about probability distributions and Shannon entropy), while Kleene's theorem is a computability-theoretic result (about codes and recursive functions). The two use no shared mathematical machinery, and ZP-C keeps surprisal and Kolmogorov complexity as two separate arguments (ZP-C Remark R-BRIDGE). Kolmogorov complexity is itself both an information-theoretic and a computability-theoretic notion; it was not located in the pinned Mathlib as of 2026-10-04, searched as recorded in ZeroParadox/Information/Disjunctive.md.

> DA-1 Path 2 is not a missing proof step — it is a genuine framework boundary. ZP-M makes the boundary precise: on one side, the fixed points over codes and ordinals (in Lean scope). On the other side, L-INF, which neither R-M.1 nor both_fixed_points_exist links to the Kleene quine, and the step from unbounded surprisal to executing: a bridge principle of its own, which the framework does not adopt (ZP-E, DA-1 Path 2).

Key Results
snapEmbed: $c_0 \mapsto 1$, $c_1 \mapsto 0$ in $\mathbb{Z}_2$ — injective, join-to-multiply morphism. ✓
hfp_from_epsilon_zero: hfp derived from monotonicity + $\varphi(\varepsilon_0) = c_1$ alone. ✓
zpm_triangle: all three edges co-proved — ordinal snap, 2-adic convergence, type bridge. ✓
both_fixed_points_exist: Kleene and ordinal fixed points co-witnessed in one formal context. ✓
R-M.1: DA-1 Path 2 boundary made precise — it does not link L-INF to the Kleene quine, and the step from unbounded surprisal to executing is an unadopted bridge principle, not a missing proof.
Every theorem proved sorry-free. Axiom footprint: at most [propext, Classical.choice, Quot.sound], per declaration in ZP-M.
